

Python Functions – Practical Questions

Beginner to Intermediate Level (Excluding Higher Order Functions)

This worksheet contains practical lab questions focused on Python functions only. Topics include function definition, arguments, return values, default arguments, keyword arguments, variable-length arguments, recursion, and small real-world logic problems.

Section A – Basic Function Questions

- 1 Write a function **greet()** that prints *Hello, Welcome to Python*.
- 2 Write a function **display_name(name)** that prints the given name.
- 3 Write a function **add(a, b)** to return the sum of two numbers.
- 4 Write a function **is_even(num)** that returns whether a number is even or odd.
- 5 Write a function **square(n)** that returns the square of a number.
- 6 Write a function **find_largest(a, b)** to return the larger number.
- 7 Write a function **check_positive(num)** to check whether a number is positive, negative, or zero.
- 8 Write a function **multiply(a, b)** and call it using user input.
- 9 Write a function **full_name(first, last)** that returns the full name.
- 10 Write a function **calculate_area(length, width)** to return the area of a rectangle.

Section B – Functions with Return Values

- 1 Write a function to return the cube of a number.
- 2 Write a function to return the reverse of a string.
- 3 Write a function to count the number of vowels in a given string.
- 4 Write a function to return the factorial of a number.
- 5 Write a function to return the sum of digits of a number.
- 6 Write a function to return the smallest element in a list.
- 7 Write a function to return the second largest number in a list.
- 8 Write a function to return the number of words in a sentence.
- 9 Write a function to return **True** if a number is a palindrome, otherwise **False**.
- 10 Write a function to return the common elements from two lists.

Section C – Arguments Practice

- 1 Write a function **student_info(name, age, course)** and call it using positional arguments.

- 2 Write the same function and call it using keyword arguments.
- 3 Write a function **power(base, exp=2)** that returns the power of a number using a default argument.
- 4 Write a function **discount(price, percent=10)** to calculate discounted price.
- 5 Write a function **introduce(name, city='Kochi')** and test it with and without the city argument.
- 6 Write a function that accepts any number of values using ***args** and returns their sum.
- 7 Write a function that accepts any number of values using ***args** and returns the largest one.
- 8 Write a function that accepts user details using ****kwargs** and prints each key-value pair.
- 9 Write a function **bill(item, quantity, price)** and calculate total bill.
- 10 Write a function **marks_total(*marks)** that returns total and average marks.

Section D – String-Based Function Problems

- 1 Write a function to check whether a string is a palindrome.
- 2 Write a function to count uppercase and lowercase letters in a string.
- 3 Write a function to remove all spaces from a string.
- 4 Write a function to find the frequency of a given character in a string.
- 5 Write a function to convert the first letter of each word to uppercase.
- 6 Write a function to replace all vowels in a string with '*'.
- 7 Write a function to check whether two strings are anagrams.
- 8 Write a function to return the longest word in a sentence.
- 9 Write a function to count how many times a word appears in a sentence.
- 10 Write a function to split a sentence and return only words with length greater than 4.

Section E – List-Based Function Problems

- 1 Write a function to return only even numbers from a list.
- 2 Write a function to return only prime numbers from a list.
- 3 Write a function to remove duplicates from a list without changing the order.
- 4 Write a function to return the sum of all odd numbers in a list.
- 5 Write a function to find the index positions of a given element in a list.
- 6 Write a function to merge two lists and return a sorted result.
- 7 Write a function to count positive and negative numbers in a list.
- 8 Write a function to return the maximum difference between two elements in a list.

- 9 Write a function to rotate a list to the right by one position.
- 10 Write a function to return a new list containing squares of all elements.

Section F – Number Logic Function Problems

- 1 Write a function to check whether a number is prime.
- 2 Write a function to generate the Fibonacci series up to n terms.
- 3 Write a function to check whether a number is an Armstrong number.
- 4 Write a function to return all divisors of a number.
- 5 Write a function to find the GCD of two numbers.
- 6 Write a function to find the LCM of two numbers.
- 7 Write a function to convert a decimal number to binary.
- 8 Write a function to count the number of digits in an integer.
- 9 Write a function to check whether a number is a perfect number.
- 10 Write a function to return the multiplication table of a number.

Section G – Recursion (Without Higher Order Functions)

- 1 Write a recursive function to find the factorial of a number.
- 2 Write a recursive function to calculate the sum of first n natural numbers.
- 3 Write a recursive function to print numbers from 1 to n.
- 4 Write a recursive function to reverse a string.
- 5 Write a recursive function to find the nth Fibonacci number.
- 6 Write a recursive function to calculate the power of a number.
- 7 Write a recursive function to count digits in a number.
- 8 Write a recursive function to check whether a string is a palindrome.
- 9 Write a recursive function to find the sum of digits of a number.
- 10 Write a recursive function to find the product of elements in a list.

Section H – Mini Real-World Function Tasks

- 1 Write a function to calculate electricity bill based on units consumed.
- 2 Write a function to calculate employee salary after bonus deduction/tax.
- 3 Write a function to calculate student grade from marks.

- 4 Write a function to check login using username and password.
- 5 Write a function to generate a simple receipt for purchased items.
- 6 Write a function to calculate simple interest and compound interest.
- 7 Write a function to validate whether a mobile number is valid.
- 8 Write a function to check password strength.
- 9 Write a function to calculate age from birth year.
- 10 Write a function to simulate an ATM withdrawal check (balance vs withdrawal amount).

Section I – Lab Exam Style Questions

- 1 Write a menu-driven program using functions for basic calculator operations.
- 2 Write a program with separate functions to input, process, and display student details.
- 3 Write a function to take a list of numbers and return both the largest and smallest.
- 4 Write a function to accept a sentence and return the number of vowels, consonants, and spaces.
- 5 Write a function to accept marks of 5 subjects and return total, average, and grade.
- 6 Write a function to accept a number and print whether it is prime, palindrome, and Armstrong.
- 7 Write a function to manage a simple contact list (add, search, display).
- 8 Write a function-based program to count how many even and odd numbers are in a list.
- 9 Write a function to simulate a shopping cart total with discount calculation.
- 10 Write a function to accept a matrix (list of lists) and print row-wise sums.

Bonus Challenge Questions

- 1 Create a library of utility functions for numbers, strings, and lists in one Python file.
- 2 Build a student result system using functions only.
- 3 Create a bank menu program using functions for deposit, withdraw, and balance check.
- 4 Create a quiz game where each task is handled by a separate function.
- 5 Build a small text-based calculator app using reusable functions.

Tip for Students: Try solving each question first without looking at old code. Focus on function design, input handling, return values, and clean logic.